

CUSTOMER SEGMENTATION & CHURN PATTERN ANALYTICS IN EUROPEAN BANKING

Abstract / Executive Summary

Customer churn is one of the major challenges in the banking sector because losing customers affects profitability, customer retention, and business growth. Banks collect large amounts of customer data, but identifying which customer groups are more likely to leave remains difficult.

This project focuses on analyzing customer churn patterns in European banking using customer segmentation and exploratory data analysis (EDA). The analysis was conducted based on customer demographics, geography, financial profile, and account activity. Key Performance Indicators (KPIs) were also calculated to measure churn among different customer segments.

Additionally, a Streamlit dashboard was developed to provide an interactive platform for churn analysis through visualizations and dynamic KPI tracking. The findings help understand churn behavior and support better customer retention strategies in banking.

INTRODUCTION

Customer churn refers to the situation where customers stop using a company's products or services. In the banking sector, customer churn is considered a major business problem because retaining existing customers is often less expensive than acquiring new ones. When customers leave a bank, it may result in reduced profits, revenue instability, and increased customer acquisition costs.

European banks manage large volumes of customer information related to demographics, financial status, account activity, and customer engagement. However, many banks struggle to understand how churn differs among various customer groups. In many cases, banks only focus on overall churn rates instead of identifying which specific customer segments are more likely to leave.

This project focuses on customer segmentation and churn pattern analytics in European banking. The project analyzes customer churn based on geography, age, tenure, credit score, account balance, and customer activity. Through systematic analysis, the project aims to identify customer groups that are at greater risk of churn.

Exploratory Data Analysis (EDA), customer segmentation techniques, and Key Performance Indicators (KPIs) were used to understand churn behavior. An interactive Streamlit dashboard was also developed to improve the visualization and accessibility of churn insights.

The project helps decision-makers understand churn trends more clearly and supports data-driven customer retention strategies.

Background of the Study

Customer retention has become an important focus area in the banking industry. Due to increasing competition among financial institutions, customers can easily switch from one bank to another if their expectations are not met.

Customer churn causes multiple challenges for banks such as reduced customer lifetime value, lower profitability, and increased marketing costs. Many banks track churn percentages but often lack detailed analysis regarding which customer groups contribute most to churn.

Different customer characteristics such as age, geography, financial stability, and account activity can influence customer behavior. Understanding these factors helps banks identify high-risk customers and implement targeted retention strategies.

This project was developed to address the need for customer segmentation-driven churn analytics in European banking. By analyzing customer profiles and churn behavior together, the project helps in identifying patterns that may not be visible through basic churn measurement alone.

Problem Statement

Despite having access to rich customer-level data, banks face challenges in identifying high-risk customer groups and understanding churn differences among various customer segments.

Many banking institutions rely on general churn measurements without detailed segmentation analysis, making customer retention strategies less efficient and reactive.

The major problem addressed in this project is the lack of structured analytical insights regarding:

- Which customer groups are most likely to churn
- How churn differs across countries and demographics

- Whether financially valuable customers are leaving the bank
- How customer engagement and tenure influence churn behavior

Without proper churn analysis, banks may fail to identify important customer risks and opportunities for retention.

Purpose of the Project

The main purpose of this project is to analyze customer churn patterns in European banking through customer segmentation and data analytics techniques.

The project aims to identify churn behavior across different customer groups and understand how demographic, geographical, and financial characteristics influence customer retention.

Additionally, the project seeks to transform raw banking data into meaningful business insights that can support better strategic planning and customer retention efforts.

Project Objectives

Primary Objectives

The primary objectives of this project are:

1. To measure the overall customer churn rate in European banking.
2. To identify customer churn distribution across different customer segments.
3. To compare churn behavior across geographical regions.

Secondary Objectives

The secondary objectives of this project are:

1. To analyze churn among high-value customers.
 2. To evaluate customer engagement and tenure patterns.
 3. To support strategic planning and customer retention decisions using data-driven insights.
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Scope of the Study

The scope of this project is limited to customer churn analysis within European banking using the available customer dataset.

The study focuses on:

- Customer segmentation analysis
- Geography-wise churn comparison
- Age and tenure churn analysis
- Credit score and balance segmentation
- Customer engagement analysis
- High-value customer churn exploration

The study does not include real-time banking transactions or predictive machine learning implementation. Instead, the focus remains on churn analytics, segmentation, and visualization.

Significance of the Study

This study is important because it helps understand customer churn behavior in a structured way. Instead of only measuring churn rates, the study identifies specific customer groups that may require additional attention.

The findings of this study can help banks:

- Improve customer retention strategies
- Identify high-risk customer segments
- Understand geography-based churn differences
- Reduce financial loss caused by customer churn
- Improve customer engagement practices

Therefore, this project provides practical value for decision-making in retail banking through data analytics and visualization.

DATASET DESCRIPTION AND METHODOLOGY

Dataset Description

The dataset used in this project contains customer-level information from European banking institutions. It includes demographic, financial, and customer engagement-related variables that help in analyzing customer churn behavior.

The dataset was used to understand how churn varies across different customer segments and to identify factors that may influence customer attrition.

Dataset Variables

Column Name	Description
CustomerId	Unique customer identification number
Surname	Customer surname
CreditScore	Customer creditworthiness score
Geography	Customer location (France, Germany, Spain)
Gender	Customer gender
Age	Customer age
Tenure	Number of years with the bank
Balance	Customer account balance
NumOfProducts	Number of banking products used
HasCrCard	Credit card ownership
IsActiveMember	Customer activity status
EstimatedSalary	Estimated annual salary
Exited	Customer churn status (Target Variable)

Target Variable:

The variable **Exited** was considered as the target variable for churn analysis.

- **0 = Retained Customer**
- **1 = Churned Customer**

Methodology

A systematic analytical methodology was followed to understand customer churn patterns in European banking.

The project was completed in the following stages:

Step 1: Data Collection

The customer banking dataset was collected and imported into Python using Pandas for analysis.

Step 2: Data Gathering and Preparation

The dataset was checked for consistency and prepared for analysis by verifying missing values, duplicate records, and data types.

Step 3: Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to identify patterns, trends, and relationships in customer churn.

The analysis included:

- Churn distribution analysis
- Geography-wise churn analysis
- Age-group churn analysis
- Credit score segmentation
- Tenure comparison
- Balance-based churn analysis
- Customer activity analysis

Step 4: KPI Analysis

Key Performance Indicators (KPIs) were calculated to measure churn performance.

KPIs included:

- Overall Churn Rate
- Segment Churn Rate
- High-Value Customer Churn Ratio
- Geographic Risk Index
- Engagement Drop Indicator

Step 5: Dashboard Development

A Streamlit web application dashboard was developed to visualize churn patterns interactively.

The dashboard included:

- Overall churn summary
- Geography-wise churn visualization
- Age & tenure churn comparison
- High-value customer churn explorer

Step 6: Findings and Recommendations

Based on EDA and KPI analysis, important findings and recommendations were developed to support customer retention strategies.

Data Gathering and Preparation

Data gathering and preparation is an important step in data analytics because raw data often contains inconsistencies or unnecessary variables.

The following steps were performed for preparing the dataset:

1. Dataset Loading

The dataset was imported into Python using the Pandas library for further analysis.

2. Dataset Inspection

The dataset structure was checked to understand:

- Number of rows and columns
- Variable names
- Data types
- Sample records

This step helped in understanding the dataset before analysis.

3. Missing Value Check

The dataset was checked for missing or null values to ensure data quality.

No significant missing values were identified in the dataset.

(Paste Screenshot: Missing Values Output Here)

4. Duplicate Value Check

Duplicate records were checked to avoid repeated customer information affecting the analysis.

The dataset did not contain major duplicate records.

(Paste Screenshot: Duplicate Check Output Here)

5. Removal of Non-Analytical Variables

The **Surname** column was not relevant for churn analysis and was excluded from analytical interpretation because it does not contribute to churn behavior.

6. Customer Segmentation Creation

To improve analysis, new customer groups were created based on customer characteristics.

The following segmentation categories were developed:

Age Groups

- Young (<30 years)
- Adult (30–45 years)
- Middle Age (46–60 years)
- Senior (60+ years)

Credit Score Groups

- Low Credit Score
- Medium Credit Score
- High Credit Score

Tenure Groups

- New Customers
- Mid-Term Customers
- Long-Term Customers

Balance Groups

- Zero Balance
- Low Balance
- High Balance

These segments helped in comparing churn behavior among different customer categories.

7. Data Validation

Binary variables such as:

- Exited
- HasCrCard
- IsActiveMember

were checked to ensure consistent values (0 and 1).

This step helped maintain analytical accuracy before Exploratory Data Analysis (EDA).

EXPLORATORY DATA ANALYSIS (EDA)

Introduction to Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand customer churn behavior and identify important trends and relationships in the banking dataset.

EDA helps in identifying hidden patterns, customer segment behavior, and churn differences across demographic and financial characteristics.

The analysis focused on:

- Overall churn distribution
- Geography-wise churn
- Gender-based churn
- Age-group churn
- Credit score churn
- Tenure churn
- Balance-based churn
- Customer activity analysis

Overall Churn Distribution

The first step of analysis was to understand the overall customer churn rate.

A churn distribution graph was created to compare churned and retained customers.

Interpretation

The analysis shows that the majority of customers were retained by the bank, while a smaller percentage of customers exited. However, the number of churned customers was still significant enough to indicate customer retention concerns.

Understanding overall churn is important because it provides a baseline for further segmentation analysis.

Geography-wise Churn Analysis

Customer churn was analyzed across different countries to understand geographical differences in churn behavior.

The analysis included:

- France
- Germany
- Spain

A geography-wise churn visualization was created to compare churn percentages across regions.

Interpretation

The analysis indicated that customer churn differed across geographical regions. Certain countries showed relatively higher churn rates compared to others, suggesting possible regional differences in customer satisfaction, financial behavior, or service expectations.

This insight can help banks implement region-specific customer retention strategies.

Gender-wise Churn Analysis

Customer churn was analyzed based on gender to examine whether churn behavior varied between male and female customers.

A comparison chart was developed for gender-based churn.

Interpretation

The analysis showed slight variation in churn behavior between male and female customers. However, gender alone may not strongly determine churn behavior and should be interpreted along with other customer characteristics.

This analysis helps understand whether customer retention strategies require gender-based targeting.

Age-group Churn Analysis

Customers were segmented into different age groups for analysis:

- Young Customers (<30 years)
- Adult Customers (30–45 years)
- Middle-Aged Customers (46–60 years)
- Senior Customers (60+ years)

Age-group churn comparison was performed using visualization techniques.

Interpretation

The analysis showed differences in churn behavior among age groups. Some customer age groups demonstrated relatively higher churn rates, indicating varying banking preferences and customer expectations.

This insight suggests that banks may need age-specific customer engagement strategies.

Credit Score Churn Analysis

Customer churn was analyzed based on credit score segmentation.

Customers were grouped into:

- Low Credit Score
- Medium Credit Score
- High Credit Score

Interpretation

Based on analysis, customers with low credit scores showed slightly higher churn compared to medium and high credit score customers.

This indicates that financial stability may influence customer retention and churn behavior.

Tenure Churn Analysis

Customer churn was analyzed based on the number of years customers remained with the bank.

Customers were categorized into:

- New Customers
- Mid-Term Customers
- Long-Term Customers

Interpretation

The analysis showed only small differences in churn rates among tenure groups. New customers displayed slightly higher churn compared to long-term customers.

This suggests that improving customer onboarding and early engagement may help reduce churn.

Balance-based Churn Analysis

Customer churn was analyzed according to account balance.

Customers were grouped into:

- Zero Balance
- Low Balance
- High Balance

Interpretation

High-balance customers showed relatively higher churn compared to low-balance customers. This finding is important because losing financially valuable customers can create greater business risk for banks.

Therefore, premium customer retention strategies may be necessary.

Customer Activity Analysis

Customer activity was analyzed using the **IsActiveMember** variable to understand the relationship between account activity and churn.

The analysis compared:

- Active Customers

- Inactive Customers

Interpretation

Inactive customers demonstrated higher churn compared to active customers. This indicates that customer engagement plays an important role in retention.

Banks may improve retention by increasing customer interaction and engagement programs.

Summary of EDA Findings

The Exploratory Data Analysis revealed important churn patterns across customer demographics, geography, financial profile, and engagement level.

Major findings include:

- Churn varied across geographical regions.
- Low credit score customers showed slightly higher churn.
- New customers demonstrated relatively higher churn.
- High-balance customers showed increased churn risk.
- Inactive customers were more likely to churn.

The EDA process helped identify customer segments that may require focused retention strategies

KPI ANALYSIS

Introduction to KPI Analysis

Key Performance Indicators (KPIs) were calculated to measure customer churn performance and identify high-risk customer segments in European banking.

The KPI analysis helped evaluate customer retention, churn distribution, customer engagement, and regional churn exposure.

The following KPIs were analyzed:

- Overall Churn Rate
- Segment Churn Rate
- High-Value Customer Churn Ratio
- Geographic Risk Index

- Engagement Drop Indicator
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Overall Churn Rate

Definition

Overall Churn Rate refers to the percentage of customers who exited the bank out of the total customer base.

Interpretation

The Overall Churn Rate provides a general understanding of customer retention performance.

A higher churn rate indicates increased customer attrition and may negatively affect bank profitability and long-term customer relationships.

This KPI serves as the foundation for understanding overall customer retention behavior.

Segment Churn Rate

Definition

Segment Churn Rate measures churn percentage across different customer groups such as:

- Credit Score Groups
- Tenure Groups
- Balance Groups
- Age Groups

This KPI helps identify which customer categories are more likely to leave the bank.

Credit Score Churn Interpretation

Based on the analysis:

- Low Credit Score Customers → **23.64%**
- Medium Credit Score Customers → **20.29%**
- High Credit Score Customers → **19.87%**

Interpretation

Customers with low credit scores showed relatively higher churn compared to medium and high credit score customers.

This suggests that financially weaker customers may be more likely to discontinue banking services.

Banks may improve retention among these customers through better financial support programs or customized banking services.

Tenure Group Churn Interpretation

Based on the analysis:

- New Customers → **20.89%**
- Mid-Term Customers → **19.64%**
- Long-Term Customers → **20.45%**

Interpretation

The difference in churn rates among tenure groups was relatively small.

However, new customers showed slightly higher churn, indicating that early customer experience and onboarding may influence customer retention.

Banks may improve retention by focusing on customer satisfaction during the initial relationship period.

Balance Group Churn Interpretation

Based on the analysis:

- High Balance Customers → **24.98%**
- Low Balance Customers → **20.82%**
- Zero Balance Customers → **13.82%**

Interpretation

High-balance customers showed the highest churn rate among all balance groups.

This finding is important because losing financially valuable customers may result in significant revenue loss for banks.

Therefore, banks should prioritize premium customer retention and personalized banking services.

Zero-balance customers showed comparatively lower churn rates.

High-Value Customer Churn Ratio

Definition

High-Value Customer Churn Ratio measures churn among financially valuable customers, particularly customers with higher account balances.

Purpose

This KPI helps identify revenue risk caused by churn among premium customers.

Interpretation

The analysis showed that financially valuable customers demonstrated notable churn behavior.

Since premium customers contribute more to bank profitability, losing these customers may create higher financial impact.

Banks should implement loyalty programs, personalized services, and relationship management strategies to retain high-value customers.

Geographic Risk Index

Definition

Geographic Risk Index evaluates churn exposure across different countries.

The analysis compared churn among:

- France
- Germany
- Spain

Interpretation

Geography-wise churn analysis revealed differences in customer retention across regions.

Certain countries showed relatively higher churn rates, indicating the need for region-specific retention strategies.

This KPI helps banks identify areas requiring additional customer engagement and service improvements.

Engagement Drop Indicator

Definition

Engagement Drop Indicator measures the relationship between customer activity and churn.

The KPI compares:

- Active Customers
- Inactive Customers

Interpretation

Inactive customers showed relatively higher churn compared to active customers.

This suggests that customer engagement plays an important role in retention.

Banks can reduce churn by improving customer interaction, communication, and service engagement.

Summary of KPI Findings

The KPI analysis provided valuable insights into customer churn behavior across different customer groups.

Major findings include:

- Low credit score customers demonstrated higher churn.
- New customers showed slightly greater churn risk.
- High-balance customers experienced the highest churn.
- Customer activity strongly influenced retention.
- Churn patterns varied across geographical regions.

The KPI analysis helped identify high-risk customer segments and supported better strategic planning for customer retention.

STREAMLIT WEB APPLICATION DASHBOARD

Introduction to Streamlit Dashboard

To improve customer churn analysis and make insights more interactive, a Streamlit web application dashboard was developed.

The dashboard provides a user-friendly platform for analyzing customer churn patterns across different customer segments. It helps users visualize churn behavior through graphs, KPIs, and interactive filtering options.

The dashboard was developed to simplify customer churn analysis and improve understanding of customer behavior in European banking.

Dashboard Objectives

The Streamlit dashboard was developed to achieve the following objectives:

- Provide an overview of customer churn
 - Compare churn across geographical regions
 - Analyze churn among different customer age groups and tenure categories
 - Identify churn among high-value customers
 - Enable interactive customer segmentation analysis
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Core Dashboard Modules

The dashboard consists of four major modules.

1. Overall Churn Summary

The Overall Churn Summary module provides a complete overview of customer churn performance.

This module includes:

- Total customers
- Churned customers
- Retained customers
- Overall churn rate

KPI indicators and charts were implemented to provide quick understanding of customer churn performance.

Interpretation

This module helps understand the overall customer retention status and the percentage of customers leaving the bank.

A higher churn percentage may indicate the need for stronger customer retention strategies.

2. Geography-wise Churn Visualization

This module analyzes customer churn across different European countries.

The following countries were included:

- France
- Germany
- Spain

Interactive visualizations were developed to compare churn behavior among geographical regions.

Interpretation

The analysis showed variation in churn rates among countries, indicating regional differences in customer behavior.

This helps banks identify high-risk regions and implement location-specific retention strategies.

3. Age & Tenure Churn Comparison

This module analyzes churn based on customer age groups and tenure with the bank.

The following categories were analyzed:

Age Groups

- Young Customers
- Adult Customers
- Middle-Aged Customers
- Senior Customers

Tenure Groups

- New Customers
- Mid-Term Customers

- Long-Term Customers

Bar charts were used to compare churn across these groups.

Interpretation

The analysis helps identify which customer age groups and relationship stages demonstrate relatively higher churn behavior.

This supports better customer engagement planning.

4. High-Value Customer Churn Explorer

This module focuses on churn among financially valuable customers.

The analysis included:

- High-value customer churn ratio
- Customer balance analysis
- Active vs inactive customer comparison

Interpretation

High-value customer churn is important because losing financially valuable customers may result in higher business loss.

This module helps banks focus retention efforts on premium customers.

User Capabilities

The Streamlit dashboard provides interactive user capabilities to improve analytical flexibility.

Segment Filters

Users can apply filters to analyze churn for specific customer groups, particularly geography-based filtering.

This helps in focused customer analysis.

Dynamic KPI Updates

KPI indicators automatically update based on selected customer segments and filters.

This improves real-time analytical understanding.

Drill-Down Views

The dashboard enables detailed analysis by moving from overall churn to segment-level churn analysis.

Users can explore:

- Geography-wise churn
 - Age-group churn
 - Tenure-based churn
 - High-value customer churn
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Advantages of the Dashboard

The Streamlit dashboard provides several advantages:

- Easy understanding of customer churn
 - Interactive data visualization
 - Quick identification of high-risk customer groups
 - Improved business decision support
 - Better customer retention planning
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Summary

The Streamlit web application successfully transformed customer churn analysis into an interactive and user-friendly system.

By integrating KPIs, charts, filters, and drill-down views, the dashboard improved understanding of customer churn patterns across demographics, geography, and financial characteristics.

FINDINGS, RECOMMENDATIONS, AND CONCLUSION

Major Findings of the Study

Based on Exploratory Data Analysis (EDA), KPI analysis, and dashboard visualization, several important findings were identified regarding customer churn in European banking.

Major Findings

1. **Customer churn remains a significant concern in European banking.**

Although many customers were retained, a noticeable percentage of customers exited the bank, indicating the importance of customer retention strategies.

2. **Geographical differences in churn were observed.**

Customer churn behavior varied across France, Germany, and Spain, suggesting that customer expectations and banking experiences may differ by region.

3. **Low credit score customers showed relatively higher churn.**

Customers with lower credit scores demonstrated slightly higher churn rates compared to medium and high credit score customers.

4. **New customers showed slightly higher churn behavior.**

Customers in the early stage of their banking relationship experienced comparatively higher churn, indicating the importance of customer onboarding and early engagement.

5. **High-balance customers showed the highest churn rate.**

Financially valuable customers were more likely to churn compared to low-balance customers, increasing potential revenue risk for banks.

6. **Inactive customers showed greater churn behavior.**

Customer activity had a strong relationship with retention, where inactive customers were more likely to leave the bank.

7. **Customer segmentation improved churn understanding.**

Analyzing churn through customer segments helped identify high-risk customer groups more effectively than using overall churn rates alone.

Recommendations

Based on the findings of the study, the following recommendations are suggested to improve customer retention in European banking.

1. Improve Customer Engagement

Banks should increase customer engagement activities to reduce churn among inactive customers.

This can include:

- Personalized communication
- Customer loyalty programs
- Digital banking engagement

2. Focus on High-Value Customer Retention

Since high-balance customers showed higher churn, banks should prioritize retention efforts for premium customers.

Banks may offer:

- Personalized banking services
- Dedicated relationship managers
- Special financial benefits

3. Strengthen New Customer Onboarding

New customers showed slightly higher churn risk.

Banks should improve onboarding experiences through:

- Better customer support
- Welcome programs
- Early customer relationship management

4. Develop Geography-Specific Retention Strategies

Since churn varied across regions, banks should design region-specific strategies rather than applying one common retention plan.

5. Improve Services for Financially Vulnerable Customers

Customers with lower credit scores showed relatively higher churn.

Banks may reduce churn through:

- Flexible financial support
- Better loan guidance
- Customized banking services

Conclusion

Customer churn is an important challenge in the banking industry because losing customers affects profitability and long-term business growth.

This project focused on customer segmentation and churn pattern analytics in European banking using Exploratory Data Analysis (EDA), KPI analysis, and an interactive Streamlit dashboard.

The study successfully analyzed churn behavior across customer demographics, geography, financial profile, and customer engagement. Important churn patterns were identified among high-value customers, inactive customers, and different geographical regions.

The Streamlit dashboard improved data visualization and enabled interactive analysis through KPIs, filters, and drill-down views.

Overall, the project helped provide a better understanding of customer churn behavior and supported data-driven customer retention strategies in European banking.

References

1. Customer Churn Dataset used for European Banking Analysis
2. Python Libraries:
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Plotly
 - Streamlit
3. Jupyter Notebook Documentation
4. Streamlit Documentation